
Q1.Attacks on Digital Signatures

Digital signatures ensure **authentication, integrity, and non-repudiation**, but they can still be attacked in various ways.

1. Key-only Attack

The attacker only has access to the **public key** and tries to generate a valid signature without knowing the private key.

2. Known Message Attack

The attacker has access to some **valid message–signature pairs** and uses them to forge a signature for a new message.

3. Chosen Message Attack

The attacker can choose arbitrary messages and obtain their signatures, then uses this information to forge signatures for other messages.

4. Replay Attack

The attacker captures a valid signed message and **retransmits it** to gain unauthorized access or repeat a transaction.

5. Man-in-the-Middle Attack

An attacker intercepts communication and may **replace or alter public keys**, misleading users into trusting fake signatures.

6. Forgery Attack

The attacker attempts to **create a fake digital signature** without access to the private key.

7. Private Key Compromise

If the **private key is stolen**, the attacker can generate valid signatures and impersonate the owner.

8. Hash Collision Attack

The attacker finds two different messages with the **same hash value**, allowing substitution of a malicious message.

Q2.Explain digital signature.

A **Digital Signature** is a mathematical technique used to validate the authenticity and integrity of a digital document, message, or software. It is the digital equivalent of a handwritten signature or a stamped seal, but it offers far more inherent security.

A digital signature provides three main security features:

1. **Authentication:** It confirms that the message was created and sent by the claimed sender.
 2. **Integrity:** It ensures that the message has not been altered during transition.
 3. **Non-repudiation:** The sender cannot deny having sent the message once the signature is verified.
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How Digital Signatures Work

Digital signatures rely on **Public Key Cryptography** (also known as Asymmetric Cryptography), which involves a pair of keys: a **Private Key** (known only to the signer) and a **Public Key** (available to everyone).

The Signing Process (Sender's Side)

1. **Hashing:** The sender uses a hash algorithm (like SHA-256) to create a fixed-length "fingerprint" of the original message, called a **Hash Value**.
2. **Encryption:** The sender encrypts this hash value using their **Private Key**. This encrypted hash is the **Digital Signature**.
3. **Sending:** The sender attaches the digital signature to the message and sends it to the recipient.

The Verification Process (Recipient's Side)

1. **Hashing:** The recipient receives the message and runs it through the same hash algorithm to create a new hash value.

2. **Decryption:** The recipient uses the sender's **Public Key** to decrypt the digital signature. This reveals the original hash value created by the sender.
3. **Comparison:** The recipient compares the two hashes:
 - **If they match:** The signature is valid (the message is authentic and unchanged).
 - **If they don't match:** Either the message was tampered with, or the signature was created with a different private key.

Key Components

- **Hash Function:** A algorithm that turns data into a unique string of characters. Even a tiny change in the data results in a completely different hash.
- **Private Key:** A secret key used by the sender to sign the hash.
- **Public Key:** A shared key used by the receiver to verify the signature.
- **Certificate Authority (CA):** A trusted third party that issues digital certificates to verify the identity of the key owners.

Common Use Cases

- **Email Security:** Using protocols like S/MIME or PGP to sign emails.
- **Software Distribution:** Developers sign their software to prove it hasn't been modified by hackers.
- **Legal Documents:** Signing digital contracts (e.g., DocuSign).
- **Financial Transactions:** Banks use digital signatures to authorize wire transfers and online payments.



